Finite terms $(\Lambda \to 0)$

Verification summary (fresh re-check). Group I Row III matches the p^+ masters only with principal-value logarithms: absolute values added below (the one substantive correction). Group II Row I second part re-verified against the p^+ master numerically (residual $\sim 10^{-22}$). The universal endpoint-log structures ($\log\left[\frac{(1-\frac{k^+}{\vee})(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]$ and $\log(1-\frac{k^+}{\vee})$) are consistent with the exact splits $\log\frac{(\vee-k^+)A+k^+B}{\Lambda A+k^+B} = \log\frac{\vee}{k^+} + \log\frac{(1-\bar\xi)A+\bar\xi B}{B} + O(\Lambda)$ and $\log\frac{(\vee-k^+)k^+}{\vee\Lambda} = \log\frac{k^+}{\Lambda} + \log(1-\bar\xi)$, both re-verified. The Group II Row IV block coincides exactly with the finite block of the step-by-step Row-IV extraction (endpoint + azimuthal + boundary, with the subtraction S kept inside the DGLAP soft pair): correct as written. Convention caveat: the entries proportional to $\log(1-\frac{k^+}{\vee})\times\left[\frac{2}{\epsilon}+\dots\right]$ (G-I Row II first part, G-II Row IV first entry, and the analogous $\log(1-\frac{k^+}{\vee})$ blocks) carry the collinear $\frac{2}{\epsilon}$: they are the $\xi=1$ plus-prescription convention terms, absorbed into the standard $\mathcal{P}_{gg}$ if the $[0,1]$ plus distribution is used; their coefficients contain no large logarithm.

1 Group I (with c.c.)

1.1 Row I

NA

1.2 Row II

We write it in two parts:

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\int_{w,w',x,x'}\!\!e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma-\log\left(\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]\times\log\left(1-\frac{k^+}{\vee}\right)\\ &\quad\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^{b}(x)-U^{ab}(x)\,\rho^{b}(x)\Big]\\ &\quad+\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\int_{w,w',x,x'}\!\!e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left\{-\frac{67}{9}+\frac{\pi^2}{3}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right\}\times\\ &\quad\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^{b}(x)-U^{ab}(x)\,\rho^{b}(x)\Big]\end{aligned}\tag{1.1}$$

$$\begin{aligned}\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w'}\!\!e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y}\!\!e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,R^i(\mathbf{k},\mathbf{p})\\ &\quad\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{hbc}\,U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}-f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\Big]\end{aligned}\tag{1.2}$$

where

$$R^i = -2k^i\log\left(\frac{p^2}{k^2+\frac{k^+}{\vee}\,p^2}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right)\right]-\frac{p^2k^i+k^2p^i}{(k-p)^2}\,\log\left(1-\frac{k^+}{\vee}\right).\tag{1.3}$$

1.3 Row III

All logarithms below whose arguments contain the difference $(x-w)^2-(x-z)^2$ (or $(x-z)^2-\frac{k^+}{\vee}(x-w)^2$) are principal-value logarithms and carry absolute values, $\log|\cdots|$ (added below): the arguments change sign on the integration domain, and the numerical check against the p^+ masters closes (residual $\lesssim 10^{-6}$) only with $\log|\cdots|$, while the plain $\log[\cdots]$ reading leaves complex $O(0.1)$ residuals. The $\mathcal{A}_0$ term and $\log[(z-w)^2/(x-w)^2]$ have positive arguments and need no bars. The second-part logs, previously written with both numerator and denominator sign-flipped, are rewritten in the same PV form as the first part

(identical under $|\cdots|$). We write it in three parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ & \\
& -\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left|\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right| \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left|\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right| \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\}
\end{aligned}
\tag{1.4}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ & \\
& \frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right|\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right|\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right|\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right|\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right|\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right|\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\}
\end{aligned}
\tag{1.5}
$$

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-z)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\}
\end{aligned}
\tag{1.6}
$$

$$
\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right] \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right] \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right] \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\}
\end{aligned}
\tag{1.7}
$$

where:

$$\mathcal{A}_0 = \frac{1}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \left\{ \arctan\left[\frac{(x-z)^2 - (x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \right] + \arctan\left[\frac{(x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \right] \right\} \tag{1.8}$$

1.4 Row IV

$$\begin{aligned}
\frac{d^3 N_{\rm NLO}}{d^3 k} = \frac{1}{(2\pi)^3} \int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)} \Bigg\{ & \\
\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2} \right] \times \Bigg[& \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2} \\
- \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2} \Bigg] & \\
+ \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right) \frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2} \right] \Bigg\} \times & \\
\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x') \right] \left[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x) \right] &
\end{aligned} \tag{1.9}$$

1.5 Row V

First part

$$\begin{aligned}
\frac{d^3 N_{1\,\rm NLO}}{d^3 k} = & \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}} \int_0^{k^+} \frac{dp^+}{2\pi} \int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \\
& \left[\delta_{im}\,\delta_{jk} - \left[\frac{k^+}{p^+}\right]_+ \delta_{jm}\,\delta_{ik} \right] \times \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x') \right] \times \\
& \left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\} \right] \\
& - \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+} \int_0^{k^+} \frac{dq^+}{2\pi} \int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \left[\frac{1}{q^+}\right]_+ \\
& \times \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x') \right] \times \left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\} \right]
\end{aligned} \tag{1.10}$$

Second part

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\int_{k^+}^0\frac{dq^+}{2\pi}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{\left[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left\{\left[\frac{k^+}{q^+}\right]_+\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.11}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+(k^+-p^+)}{k^{+2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\,\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\right]\\
&\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\,\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\,\left[\frac{k^+}{q^+}\right]_+\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\,\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{11}{6}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.12}
$$

2 Group II

2.1 Row I

2.1.1 First part

$$\frac{d^3N_{1a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times$$
$$\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}$$
$$\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{c'b}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{2.1}$$

$$\frac{d^3N_{1b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{$$
$$\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w')^2}{(x-w')^2}\right]\times$$
$$\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]$$
$$+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}$$
$$\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{c'b}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{cb}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right] \tag{2.2}$$

2.1.2 Second part

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\Bigg\{$$
$$\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times$$
$$\Bigg[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}$$
$$+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}$$
$$+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}$$
$$-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\Bigg]\times$$
$$\left[\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]-\log\left[\frac{(x-w)^2}{(x'-w')^2}\right]\right]\Bigg\}\times$$
$$\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\Bigg] \tag{2.3}$$

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$

$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$

$$\left.+\,\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times$$

$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$

$$\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]$$

$$-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$

$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$

$$\left.+\,\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times$$

$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$

$$\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]$$

$$+\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\;-(x-w)^2\,(x'-z)^2\,(x'-w')^2]}\times$$

$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$

$$\left.+\,\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]$$

$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$

$$\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{2.4}$$

$$
\begin{aligned}
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{2.5}
$$

$$
\begin{aligned}
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} &= +\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{\left[(x-z)^2-(x-w)^2\right]}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{\left[(x'-w')^2-(x'-z)^2\right]}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{2.6}
$$

$$
\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{2.7}
$$

$$
\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]}\\
&\frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\left[\frac{1}{\vee}-\frac{1}{k^+}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{(z-w)^2\,(z-w')^2}\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}\\
&\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\,\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{2.8}
$$

2.2 Row II

NA

2.3 Row III

NA

2.4 Row IV

(checked: identical to the finite block of the Row-IV note — endpoint, azimuthal, boundary; the subtraction S sits in the DGLAP soft pair and S + boundary + BFKL = 0; no changes needed)

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^4}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\\
&\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\\
&+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\\
&\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,w,w',x',z} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2}\\
&\left[U^{ce'}(z)-U^{ce'}(x')\right]\left[U^{ce}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)
\end{aligned}
\tag{2.9}
$$

3 Group III (with c.c.)

3.1 Row I

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\
&\left.-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right] \\
&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \\
&\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y)-U^{d'b}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times \\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{3.1}
$$

3.2 Row II

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\
&\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right] \\
&-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \\
&\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{c'b}(z)\,\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{3.2}
$$

3.3 Row III

3.3.1 First part

$$
\begin{aligned}
\frac{d^3N_{1\,\rm NLO}}{d^3k} = &\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2} \\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_++\frac{1}{(p^++k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\left[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times \\
&\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \\
&+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2} \\
&\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\left[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right. \\
&\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\left[U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times \\
&\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \\
&-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{16\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{w,x',y',z,y,x}e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2} \\
&\left[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]
\end{aligned}
\tag{3.3}
$$

3.3.2 Second part

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\\
&\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)^2}\,\left[(x-z)^2-(x-w)^2\right]+\frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.\\
&\left.-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]+\frac{\delta_{k'm}}{(p^++k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2}+\frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]\\
&-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_+\!+\frac{1}{(p^++k^+)}\right)\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2}+\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]\\
&\left.-\frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2}-\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2}+\frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times\\
&\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m}{(y'-z)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\\
&\left[-\frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times\\
&\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.\\
&\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\left[f^{c'd'a}\,U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(\right.\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-z)^{k'}}{(w'-z)^2\,(x'-z)^2}\\
&\left[\frac{(x-w)^2}{(x-z)^2\,(p^++k^+)\,[p^+\,(x-z)^2+k^+\,(x-w)^2]}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times\\
&\left[f^{c'd'a}\,U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.\\
&\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{3.4}
$$

3.4 Row IV

NA

3.5 Row V

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}\,(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_+\!+\frac{1}{(p^++k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\,\Big[U^{c'e'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]\\
&+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}\,(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}\\
&\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\,\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\Big[U^{c'e'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right.\\
&\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\Big[U^{c'e'}(z)\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right]\times\\
&\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{c'e'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]
\end{aligned}
\tag{3.5}
$$

3.6 Row VI

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}\,(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_+\!+\frac{1}{(p^++k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\,\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]\\
&+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}\,(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}\\
&\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\,\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right.\\
&\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\Big[U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right]\times\\
&\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]
\end{aligned}
\tag{3.6}
$$